Jorge Sánchez Aguilar

AI Engineer | Agentic Systems · Computer Vision · Production AI

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SUMMARY

AI Engineer at Ferrovial Digital Hub building production computer vision systems for US highway infrastructure, and agentic LLM systems on my own time. NASA Space Apps 2025 Global Finalist. CS at URJC with all coursework completed and thesis in progress, graduating Sept/Oct 2026. Focus: taking AI systems from model training through deployment, with an emphasis on agentic architectures, observability and CI.

EXPERIENCE

AI Engineer Intern — Digital Hub, Ferrovial — Sept 2025 – Present

Madrid, Spain (Hybrid) · Computer Vision team — US Highway Tolling Projects

- Run the full computer vision workflow end-to-end: dataset curation from raw CCTV footage, model training on Azure ML compute, formal evaluation (F1, precision/recall, confusion matrices), and automated reporting used by the US team to support project decisions.
- Designed and built a night-time vehicle detection model from scratch, fine-tuning YOLO26-Pose for low-quality public CCTV with heavy headlight noise; selected for production over CenterNet and other keypoint alternatives evaluated by the team, at 3x faster inference and comparable accuracy.
- Designed and shipped a non-AI camera perspective-shift detector using geometric heuristics and a 3-check validation gate (2-of-3 required), detecting sub-pixel CCTV movement in real time; selected for production over ML alternatives due to speed and accuracy.
- Stack: Python, PyTorch, OpenCV, Docker, Azure ML, Azure DevOps, Azure Blob Storage, MLflow, Git.

PROJECTS

The Tenth Floor — Multi-Agent LLM System for Financial Market Analysis — March 2026 – April 2026

- Architected a 3-agent LLM pipeline (MacroAnalyst, TradeAnalyst, RiskReviewer) producing validated trade proposals across 20 assets in 4 asset classes (crypto, equities, ETFs, commodities); each agent is traced end-to-end in Langfuse for full observability of prompts, tokens, and latency.
- Deployed local inference with Qwen3-32B-AWQ on vLLM via Docker Compose with health-checked services and hardware profiles; provider-agnostic routing (LLM_BASE_URL override) allows switching between local vLLM and OpenRouter without code changes.
- Engineered a typed data layer with Pydantic v2 frozen contracts at every module boundary, SQLite with versioned migrations and full signal lifecycle tracking, and a fully-mocked unit test suite running in GitHub Actions CI with pytest, ruff, and mypy on every push.
- Built a FastAPI backend and a React 19 dashboard streaming live pipeline events over WebSocket, with manual run triggering and signal tracking views.
- Ran the pipeline as a live forward test, tracked every signal to resolution, and stopped the experiment when the sample showed no evidence of edge; results and limitations documented in the repo.

Space Atlas — NASA Space Apps Challenge 2025 — Oct 2025

Team of 5 · Madrid Winner · Global Finalist

- Built the Node.js/Express backend and deployed it to Azure App Service with a GitHub Actions CI/CD pipeline, integrating Azure OpenAI (GPT-4o) with Azure Blob Storage through a REST API that generates real-time educational explanations of Martian geological features from high-resolution imagery.

EDUCATION

B.S. Computer Science — Universidad Rey Juan Carlos — Expected Sept/Oct 2026

All coursework completed. Relevant coursework: Artificial Intelligence, Computer Vision, Advanced Data Structures, Operating Systems. Madrid, Spain.

Final thesis (in progress): computer vision system for padel shot detection.

42 Madrid — Peer-to-peer software engineering program — 2023 – 2025

Focus: C/C++ systems programming, UNIX/Linux, object-oriented design, AI/ML.

AWARDS & COMPETITIONS

- **NASA Space Apps Challenge 2025 — Madrid Winner · Global Finalist**. Advanced to global finalist round out of 11,500+ teams and 114,000+ participants worldwide.
- **Qubic × Vottun Madrid Hackathon 2025 — Top 10 Finalist**

SKILLS

AI / ML: Computer Vision, Object Detection, Pose Estimation, LLMs, Multi-Agent Systems, Prompt Engineering, vLLM, Hugging Face Transformers, NLP.

Programming: Python (PyTorch, scikit-learn, Pandas, NumPy, OpenCV, Pydantic, FastAPI), JavaScript/TypeScript (Node.js, React), C, C++, Java, SQL.

Tools & Cloud: MLOps, Docker, Git, GitHub Actions (CI/CD), REST APIs, Azure (OpenAI, ML, Blob Storage, App Service, DevOps), Langfuse, MLflow, Linux/UNIX.

Languages: English (C2 Proficiency), Spanish (Native).